Week 1

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)
- Explain the role of probability in statistics. (CLO 4)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe statistics as the process of collecting, describing, and analyzing data. (CLO 1) **2.** Explain how to judge a striking result by how likely it is to occur by chance alone. (CLO 4) **3.** Distinguish a sample from a population. (CLO 1) **4.** Explain how a non-representative sample biases an estimate. (CLO 1) **5.** Compute the mean and median of a small dataset. (CLO 2) **6.** Explain how an outlier affects the mean and the median. (CLO 2)	START HERE: Welcome; Syllabus; Textbook and Wiley Course Resources; Online Tutoring; College Policies & Student Support Services Introduction Slides: What Is Statistics For? (Day 1)	Participation: Syllabus – course tour and expectations Day-1 class data activity: collect everyone's age, then find the mean, median, mode and boxplot together; compare the class median to the U.S. median age	Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	Blackboard Zoom Respondus
7. Identify the cases and variables in a dataset presented as a table. (CLO 2) **8.** Classify a variable as categorical or quantitative. (CLO 2) **9.** Identify the explanatory and response variables in a research question. (CLO 1)	e-Text: 1.1 The Structure of Data Interactive Lecture: 1.1 The Structure of Data Slides: 1.1 The Structure of Data	Participation: Variables – identify cases, variables and variable types; Your Turn: Used Cars: Cases and Variables	Online Homework: 1.1 The Structure of Data (due Sun Aug 30, 11:59 PM) Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Week 2

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Distinguish a population from a sample. (CLO 1) **2.** Explain how statistical inference uses sample data to draw conclusions about a population. (CLO 1) **3.** Identify sampling bias and classify the sampling method (random, convenience, or volunteer) in a described study. (CLO 1) **4.** Evaluate whether a study's results can be generalized to the population. (CLO 1)	e-Text: 1.2 Sampling from a Population Interactive Lecture: 1.2 Sampling from a Population Slides: 1.2 Sampling from a Population (includes sampling video)	Participation: Variables in Statistics (Kahoot!) – Tue: review of Week 1 Participation: Sampling – spot the bias in sample scenarios; Your Turn: Name the Bias Participation: Sampling Methods (Kahoot!) – Thu: sampling review	Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM) Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Kahoot! Zoom Respondus
5. Classify a study as observational or experimental. (CLO 1) **6.** Identify a plausible confounding variable in an observational study. (CLO 1) **7.** Explain how random assignment, a control group, and blinding support cause-and-effect conclusions. (CLO 1) **8.** Distinguish random sampling from random assignment. (CLO 1)	e-Text: 1.3 Experiments and Observational Studies Interactive Lecture: 1.3 Experiments and Observational Studies Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment; Your Turn: Study Design and Scope of Inference (coffee and depression; vitamin D experiment)	Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM) Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Week 3

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct frequency and relative frequency tables for a categorical variable. (CLO 2) **2.** Compute a sample proportion $\hat{p}$. (CLO 2) **3.** Distinguish the parameter p from the statistic $\hat{p}$. (CLO 2) **4.** Compute overall and conditional proportions from a two-way table, choosing the correct denominator. (CLO 2) **5.** Compute and interpret a difference in proportions. (CLO 2) **6.** Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CLO 2)	e-Text: 2.1 Categorical Variables Interactive Lecture: 2.1 Categorical Variables Slides: 2.1 Categorical Variables	Participation: Frequency and Relative Frequency Participation: Finding Proportions from Two-Way Tables – Your Turn: Two-Way Table Proportions (pet ownership by housing)	Online Homework: 2.1 Categorical Variables (due Sun Sep 13, 11:59 PM) Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus
7. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CLO 2) **8.** Compute the mean and median of a dataset. (CLO 2) **9.** Explain how skewness and outliers affect the mean and the median (resistance). (CLO 2) **10.** Interpret the standard deviation as the typical distance from the mean. (CLO 2) **11.** Apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CLO 2)	e-Text: 2.2 One Quantitative Variable Interactive Lecture: 2.2 One Quantitative Variable Slides: 2.2 One Quantitative Variable	Participation: Quantitative Variables – shape, mean vs. median; Your Turn: Mean vs. Median with an Outlier (commute times) Estimation activity: estimate the mean and SD of a dotplot without calculating (95% rule)	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 13, 11:59 PM) Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Week 4

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)
- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the sample standard deviation of a small dataset by hand. (CLO 2) **2.** Compute a z-score for a data value. (CLO 2) **3.** Use z-scores to compare values measured on different scales or in different groups. (CLO 2) **4.** Compute the five-number summary, range, and IQR of a dataset. (CLO 2) **5.** Apply the 1.5×IQR rule to identify outliers. (CLO 2) **6.** Construct a boxplot, marking any outliers. (CLO 2)	e-Text: 2.3 One Quantitative Variable: Using Percentiles Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles Slides: 2.3 One Quantitative Variable: Using Percentiles	Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2 Participation: Using Percentiles – Your Turn: Comparing Across Groups (heights of men and women with z-scores) Participation: Box Plots and Outliers – five-number summary, 1.5×IQR rule	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM) Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Kahoot! Zoom Respondus
7. Compare a quantitative variable across groups using side-by-side boxplots. (CLO 2) **8.** Compute and interpret a difference in means. (CLO 2) **9.** Describe the direction, form, and strength of the relationship in a scatterplot. (CLO 7) **10.** Estimate the correlation r from a scatterplot. (CLO 7) **11.** Use technology to compute the correlation r. (CLO 7) **12.** Explain how outliers, nonlinear patterns, and the absence of causation limit the interpretation of r. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Examples: tea vs. coffee boxplots and difference in means; NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier Your Turn: Tea vs. Coffee: Descriptive Statistics (difference in means; can we conclude causation?)	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM) Take-Home Quiz 3 (due Tue Oct 6, 11:59 PM) – covers 2.4–2.5 (objectives 9–12) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS Blackboard Zoom Respondus

Week 5

Course Level Objectives:

- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Interpret the slope of a regression line in context with correct units. (CLO 7) **2.** Interpret the intercept of a regression line in context and evaluate whether it is meaningful. (CLO 7) **3.** Use a regression equation to predict a response value. (CLO 7) **4.** Compute and interpret a residual. (CLO 7) **5.** Locate positive and negative residuals on a scatterplot. (CLO 7) **6.** Use technology to find the least-squares regression line. (CLO 7) **7.** Explain what “least squares” means. (CLO 7) **8.** Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, and influential outliers. (CLO 7) **9.** Explain why a strong regression relationship does not by itself establish cause and effect. (CLO 7)	e-Text: 2.5 Two Quantitative Variables: Linear Regression Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) Slides: 2.5 Two Quantitative Variables: Linear Regression	Participation: Two Variable Relationships – Tue: correlation review Participation: Regression Line – Thu: baseball game length and commute-time examples; Your Turn: Putting It All Together (used-car price vs. age: slope, intercept, prediction, residual)	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM) Take-Home Quiz 3 (due Tue Oct 6, 11:59 PM) – covers 2.4–2.5 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Week 6

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. List the sample space for a chance process with equally likely outcomes. (CLO 3) **2.** Compute probabilities by counting equally likely outcomes. (CLO 3) **3.** Apply the complement rule. (CLO 3) **4.** Apply the addition rule to find P(A or B) from a two-way table. (CLO 3) **5.** Compute conditional probabilities from a two-way table. (CLO 3) **6.** Apply the multiplication rule for draws with and without replacement. (CLO 3) **7.** Distinguish disjoint events from independent events. (CLO 3)	e-Text: P.1 Probability Rules Interactive Lecture: P.1 Probability Rules Slides: P.1 Probability Rules	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement; Your Turn: Probabilities from a Table (job by enrollment status)	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus
8. Classify a numerical summary as a parameter or a statistic. (CLO 5) **9.** Use the correct notation for parameters (μ, σ, p, ρ) and statistics ($\bar{x}$, s, $\hat{p}$, r). (CLO 5) **10.** Describe how a sampling distribution is built from repeated samples and what each dot represents. (CLO 5) **11.** Describe the center and shape of a sampling distribution. (CLO 5) **12.** Explain the standard error as the variability of a statistic. (CLO 5) **13.** Predict how the sample size affects the standard error. (CLO 5)	e-Text: 3.1 Sampling Distributions Interactive Lecture: 3.1 Sampling Distributions Slides: 3.1 Sampling Distributions	Your Turn: Parameter or Statistic? (name each number as a parameter or statistic and give its symbol)	Online Homework: 3.1 Sampling Distributions (due Sun Oct 11, 11:59 PM) Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers 3.1–3.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Also this week: Test 1 – Fri Oct 2 – Sun Oct 4, 11:59 PM (proctored, 120 min). Covers Weeks 1–5.

Week 7

Course Level Objectives:

- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct a 95% confidence interval as statistic ± 2·SE, given the statistic and its standard error. (CLO 6) **2.** Interpret a confidence interval in context. (CLO 6) **3.** Explain the confidence level as the long-run proportion of intervals that capture the parameter. (CLO 4, CLO 6) **4.** Identify common misinterpretations of a confidence interval. (CLO 6)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals Slides: 3.2 Understanding and Interpreting Confidence Intervals	Participation: Sampling Distribution (Kahoot!) – Tue: review of 3.1 Participation: Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations; Your Turn: Do These CIs Capture p?	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM) Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers 3.1–3.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS Blackboard Kahoot! Zoom Respondus
5. Describe how a bootstrap sample and a bootstrap distribution are created. (CLO 5, CLO 6) **6.** Explain why a bootstrap distribution is centered at the sample statistic. (CLO 5) **7.** Use StatKey to generate a bootstrap distribution. (CLO 5, CLO 6) **8.** Estimate the standard error as the standard deviation of the bootstrap distribution. (CLO 5, CLO 6) **9.** Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)	e-Text: 3.3 Constructing Bootstrap Confidence Intervals Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator)	Participation: Bootstrap Confidence Intervals – Your Turn: Bootstrap a Proportion (32 of 50 prefer Brand A): build the bootstrap distribution in StatKey, read off SE, form the CI	Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM) Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers 3.1–3.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS Blackboard StatKey Respondus

Also this week: Test 1 Retake – due Sun Oct 11, 11:59 PM (optional; proctored, 120 min; higher score counts)

Week 8

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval. (CLO 6) **2.** Construct a percentile bootstrap confidence interval with StatKey. (CLO 6) **3.** Predict how changing the confidence level affects the width of an interval. (CLO 6) **4.** Predict how changing the sample size affects the width of an interval. (CLO 6) **5.** Evaluate whether a bootstrap distribution is symmetric and bell-shaped enough to trust a bootstrap interval. (CLO 6) **6.** Explain why a bootstrap interval cannot correct for a non-representative sample. (CLO 1, CLO 6)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	Participation: Confidence Intervals Using Percentiles – StatKey: 90/95/99% intervals for the same data; compare the SE and percentile methods; Your Turn: Percentile CIs (124 of 200 students work)	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
7. Identify the parameter of interest in a research question. (CLO 8) **8.** Write null and alternative hypotheses in symbols and in words. (CLO 8) **9.** Classify a test as right-tailed, left-tailed, or two-tailed. (CLO 8) **10.** Explain the logic of a hypothesis test: assume H_0 is true and ask whether the sample data are surprising. (CLO 8)	e-Text: 4.1 Introducing Hypothesis Tests Interactive Lecture: 4.1 Introducing Hypothesis Tests Slides: 4.1 Introducing Hypothesis Tests	Participation: Introduction to Hypothesis Tests – rock paper scissors (9 of 11 wins): define the parameter, state H_0 and H_a, choose the tail; Your Turn: Write the Hypotheses (airline on-time rate, SAT prep, and more)	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Week 9

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how a randomization distribution is generated assuming H_0 is true. (CLO 5, CLO 8) **2.** Identify the center of a randomization distribution as the null value. (CLO 5) **3.** Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5) **4.** Compute a p-value from a randomization distribution as the proportion of simulated statistics at least as extreme as the observed statistic. (CLO 3, CLO 8) **5.** Determine the correct tail(s) for a p-value from the alternative hypothesis. (CLO 8) **6.** Interpret a p-value in context. (CLO 4, CLO 8)	e-Text: 4.2 Measuring Evidence with P-Values Interactive Lecture: 4.2 Measuring Evidence with P-Values Slides: 4.2 Measuring Evidence with P-Values	Participation: P-Values – StatKey: dolphin study (16 flips); count how many simulated statistics are as extreme as observed; Your Turn: Tea and Immunity in StatKey (re-randomize tea vs. coffee, find the p-value)	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
7. Relate the size of a p-value to the strength of evidence against H_0. (CLO 8) **8.** Compare a p-value to the significance level α to decide whether to reject H_0. (CLO 8) **9.** Write a two-step conclusion in context that avoids saying "accept H_0". (CLO 8)	e-Text: 4.3 Determining Statistical Significance Interactive Lecture: 4.3 Determining Statistical Significance Slides: 4.3 Determining Statistical Significance (includes "What is a p-value?" video)	Participation: Determining Statistical Significance – resveratrol examples (p = 0.04, 0.12, 0.0003): strength of evidence, decision at α, conclusion in context; Your Turn: Decide and Conclude	Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Week 10

Course Level Objectives:

- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe Type I and Type II errors in the context of a study. (CLO 8) **2.** Relate the significance level α to the probability of a Type I error. (CLO 4, CLO 8) **3.** Explain the trade-off between Type I and Type II errors and how sample size affects it. (CLO 8) **4.** Explain how multiple testing and publication bias produce false positives. (CLO 8) **5.** Distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions; Your Turn: Errors and Multiple Tests (ambulance response times)	Online Homework: 4.4 A Closer Look at Testing (due Sun Nov 1, 11:59 PM) Take-Home Quiz 6 (due Tue Nov 3, 11:59 PM) – covers 4.4–4.5 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus
6. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping). (CLO 8) **7.** Identify the centers of the bootstrap and randomization distributions for given data. (CLO 5) **8.** Use a 95% confidence interval to reach a two-tailed test conclusion at α = 0.05. (CLO 6, CLO 8) **9.** Determine whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples and where each distribution is centered; Your Turn: CI or Test? (smart-speaker ownership)	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM) Take-Home Quiz 6 (due Tue Nov 3, 11:59 PM) – covers 4.4–4.5 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Also this week: Test 2 – Fri Oct 30 – Sun Nov 1 (proctored, 120 min). Covers Weeks 6–9.

Week 11

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Explain when the Central Limit Theorem justifies replacing a randomization distribution with a normal distribution N(null value, SE). (CLO 5) **2.** Compute a standardized z statistic. (CLO 8) **3.** Determine a p-value as a right-tail, left-tail, or two-tail area under the standard normal curve. (CLO 3, CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value; Your Turn: z-Tests (right-, left-, and two-tailed)	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 8, 11:59 PM) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Respondus
4. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5) **5.** Construct a confidence interval for a population proportion using z* for 90%, 95%, or 99% confidence. (CLO 6) **6.** Interpret a confidence interval for a proportion in context. (CLO 6) **7.** Perform a one-proportion z test and state the conclusion in context. (CLO 8) **8.** Explain why the confidence-interval standard error uses $\hat{p}$ while the test uses p_0. (CLO 8)	e-Text: 6.1 Inference for a Proportion (6.1-D Distribution, 6.1-CI Confidence Interval, 6.1-HT Hypothesis Test) Interactive Lecture: 6.1 Inference for a Proportion Slides: 6.1 Inference for a Proportion StatKey – Theoretical Distributions (Normal)	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs; Your Turn: Testing a Proportion (60 of 250 live on campus)	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM) Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Also this week: Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)

Week 12

Course Level Objectives:

- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Determine whether a study calls for one-proportion or difference-in-proportions inference. (CLO 5) **2.** Check the conditions for inference about a difference in proportions. (CLO 5) **3.** Construct a confidence interval for $p_1 - p_2$. (CLO 6) **4.** Interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CLO 6) **5.** Compute the pooled proportion. (CLO 8) **6.** Perform a two-proportion z test from summary data or a two-way table. (CLO 8)	e-Text: 6.3 Inference for a Difference in Proportions (6.3-D, 6.3-CI, 6.3-HT) Interactive Lecture: 6.3 Inference for a Difference in Proportions Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	Participation: Difference in Proportions – two-proportion randomization (18/30 vs. 12/30); compare simulation with the formula; Your Turn: Treatment vs Placebo (two-proportion z test from a two-way table)	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 15, 11:59 PM) Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus
7. Explain why inference for a mean uses the t distribution with df = n − 1. (CLO 5) **8.** Check the conditions for t procedures. (CLO 5) **9.** Construct and interpret a t confidence interval for a population mean. (CLO 6) **10.** Perform a one-sample t test and state the conclusion in context. (CLO 8)	e-Text: 6.2 Inference for a Mean (6.2-D, 6.2-CI, 6.2-HT) Interactive Lecture: 6.2 Inference for a Mean Slides: 6.2 Inference for a Mean StatKey – Theoretical Distributions (t)	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups; Your Turn: Testing a Mean (juice bottles underfilled?)	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM) Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Week 13

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the standard error and conservative degrees of freedom for a difference in two independent means. (CLO 5) **2.** Check the conditions for two-sample t procedures. (CLO 5) **3.** Construct and interpret a confidence interval for $\mu_1 - \mu_2$. (CLO 6) **4.** Explain how the order of the groups affects the sign of the difference. (CLO 6) **5.** Perform a two-sample t test, choosing the tail from the research question. (CLO 8)	e-Text: 6.4 Inference for a Difference in Means (6.4-D, 6.4-CI, 6.4-HT) Interactive Lecture: 6.4 Inference for a Difference in Means Slides: 6.4 Inference for a Difference in Means	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test; Your Turn: Exercise and Sleep (two-sample t test and CI)	Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus
6. Classify a study design as paired or independent samples by checking for a link between observations. (CLO 1, CLO 8) **7.** Define the difference variable d and the order of subtraction. (CLO 8) **8.** Perform a paired t test. (CLO 8) **9.** Construct and interpret a paired t confidence interval. (CLO 6) **10.** Explain why pairing reduces variability. (CLO 8)	e-Text: 6.5 Paired Difference in Means Interactive Lecture: 6.5 Paired Difference in Means Slides: 6.5 Paired Difference in Means	Participation: Paired or Independent? – Your Turn: Paired or Unpaired? (class discussion of study scenarios); define d and the order of subtraction	Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM) Take-Home Quiz 8 (due Tue Nov 24, 11:59 PM) – covers 6.5 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus

Also this week: Last day to withdraw: Thu Nov 19.

Week 14

Course Level Objectives:

- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. State the hypotheses for a chi-square test of association from a two-way table. (CLO 8) **2.** Compute the expected counts for a two-way table. (CLO 8) **3.** Check that all expected counts are at least 5. (CLO 8) **4.** Compute the chi-square statistic and its degrees of freedom, $(r-1)(c-1)$. (CLO 8) **5.** Interpret the right-tail p-value of a chi-square test. (CLO 8) **6.** Relate the chi-square test on a 2×2 table to the two-proportion z test. (CLO 8) **7.** Explain why a significant result shows association but not direction or causation. (CLO 8)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables Slides: 7.2 Testing for an Association Between Two Categorical Variables StatKey – Theoretical Distributions (χ^2)	Participation: Chi-Square Test – Your Turn: Faculty Rank and Gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom

Also this week: Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)

Week 15

Course Level Objectives:

- Describe and compute confidence intervals. (CLO 6)
- Solve linear regression and correlation problems. (CLO 7)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Identify when ANOVA is appropriate. (CLO 8) **2.** State H_0 and H_a for ANOVA in symbols and in words. (CLO 8) **3.** Explain what SSG and SSE measure. (CLO 8) **4.** Explain how the ratio of between-group to within-group variability gives evidence against H_0. (CLO 8) **5.** Complete an ANOVA table (df, MS, F) from given sums of squares. (CLO 8) **6.** Interpret the F statistic and p-value in context. (CLO 8) **7.** Check the conditions for ANOVA. (CLO 8)	e-Text: 8.1 Analysis of Variance Interactive Lecture: 8.1 Analysis of Variance Slides: 8.1 Analysis of Variance StatKey – Theoretical Distributions (F)	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide; Your Turn: Complete the ANOVA Table (fertilizer and seedling growth)	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom
8. State the simple linear model. (CLO 7) **9.** Distinguish the population slope β_1 and correlation ρ from the sample estimates b_1 and r. (CLO 7) **10.** Interpret the slope in context. (CLO 7) **11.** Use regression output to test H_0: $\beta_1 = 0$ with a t statistic on $n - 2$ degrees of freedom. (CLO 7, CLO 8) **12.** Explain why the tests for slope and for correlation give the same p-value. (CLO 7) **13.** Construct and interpret a confidence interval for the slope. (CLO 6, CLO 7) **14.** Interpret R^2 as the proportion of variability in the response explained by the model. (CLO 7) **15.** Check the conditions for regression inference with a residual plot. (CLO 7)	e-Text: 9.1 Inference for Slope and Correlation Interactive Lecture: 9.1 Inference for Slope and Correlation Slides: 9.1 Inference for Slope and Correlation	Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, R^2); Your Turn: Alcohol and Missed Classes (drinks per week vs. classes missed)	Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom

Week 16

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Select the appropriate procedure for a research question among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression. (CLO 8) **2.** Perform a complete hypothesis test, from hypotheses to a conclusion in context. (CLO 8) **3.** Construct and interpret a confidence interval, using simulation or theoretical distributions as appropriate. (CLO 6) **4.** Evaluate whether a study's design allows its conclusion to be generalized to a population. (CLO 1) **5.** Evaluate whether a study's design supports a cause-and-effect conclusion. (CLO 1)	Review Slides: Putting It All Together (with answer key and no-answers versions) All Section Slides and Interactive Lectures (Weeks 1–15) StatKey	Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI	Final Exam – Thu Dec 10 (proctored, 2 hours, cumulative; no retake) (all objectives)	Blackboard WileyPLUS StatKey Zoom Respondus

Also this week: Late WileyPLUS homework closes Tue Dec 8, 11:59 PM. A higher final exam score replaces the lowest test score.